

MOBILE PHONE DETECTOR Using LM358

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This mobile phone detector can sense the presence of an activated mobile phone from a distance of four to five metres. So it can come handy in an examination hall or meetings where mobile phones are not permitted.

The circuit can detect incoming and outgoing calls, SMSes, Internet and video transmissions even if a mobile phone is kept in silent mode. When it detects an RF signal from an activated mobile phone, its LED starts blinking and continues to blink until the signal stops. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the mobile phone detector using LM358 is shown in Fig. 2. It is built around LM358 (IC1) and npn transistor BC548 (T1).

When a mobile phone is active, it radiates RF signal that passes through nearby space. The signal contains electromagnetic RF radiation from the phone.

Capacitor C1 is used in the circuit to detect the RF signal from the mobile phone. When the mobile phone radiates energy in the form of RF signal, C1 absorbs it and passes on to the inputs of IC1. This is indicated by the flashing of LED1. Preset VR1 (2.2M) is used to vary the range of the circuit. Transistor T1 is used to amplify the signal obtained at pin 1 of IC1.

The circuit is applicable for 2G networks, GPRS and network

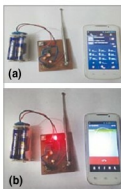


Fig. 1: Author's prototype (a) when a call is not detected and (b) when a call is detected

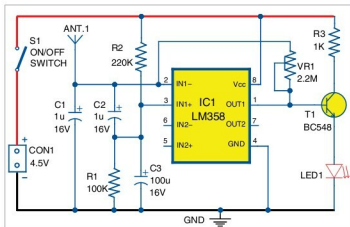


Fig. 2: Circuit diagram of the mobile phone detector

PARTS LIST

Semiconductors:

- IC1 - LM358 op-amp
- T1 - BC548 npn transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 100-kilo-ohm
- R2 - 220-kilo-ohm
- R3 - 1-kilo-ohm

Capacitors:

- C1, C2 - 1 μ F, 16V electrolytic
- C3 - 100 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- S1 - On/off switch
- ANT.1 - 15cm single-strand wire antenna

- 4.5V DC power supply

search (manual/automatic). It does not detect 3G, WCDMA and HSDPA network signals so well.

Construction and testing

An actual-size, single-side PCB layout for the mobile phone detector circuit is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable plastic box.

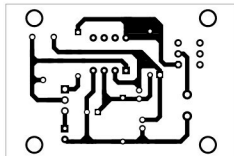


Fig. 3: Actual-size PCB layout for the mobile phone detector circuit

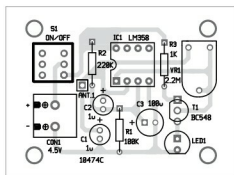


Fig. 4: Component layout of the PCB

The circuit works off a 4.5V DC power supply. **EFY**



Kumar Abhisekh is an electronics hobbyist